

Yi-Ruei Liu

Incoming PhD student at National Yang Ming Chiao Tung University (NYCU) with a Master of Computer Science from UIUC.
Focused on Computer Vision research, particularly in 3D vision and physically based rendering.

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Education

National Yang Ming Chiao Tung University , Computer Science	Hsinchu, Taiwan Sept 2026 – June 2030
University of Illinois at Urbana-Champaign , Computer Science	Illinois, US Sept 2024 – Jan 2026
National Yang Ming Chiao Tung University , Computer Science	Hsinchu, Taiwan Sept 2021 – Jan 2025

Experience

Academia Sinica , Research Assistant Led a research project development on synthesizing camera effects from real-world videos.	Taipei, Taiwan June 2025 – Sept 2025 4 months
CompPhotoLab, National Yang Ming Chiao Tung University , Research Assistant Researched and implemented algorithms for 3D shape reconstruction from single-view images under dynamic lighting.	Hsinchu, Taiwan May 2023 – June 2024 1 year 2 months

Publications

Skyfall-GS: Synthesizing Immersive 3D Urban Scenes from Satellite Imagery The first method to synthesize immersive, real-time navigable 3D urban scenes solely from multi-view satellite imagery. Jie-Ying Lee, Yi-Ruei Liu, Shr-Ruei Tsai, Wei-Cheng Chang, Chung-Ho Wu, Jiewen Chan, Zhenjun Zhao, Chieh Hubert Lin, Yu-Lun Liu skyfall-gs.jayinnn.dev (arXiv)	Oct 2025
On the Generation and Transformation of Light - A Heuristic Viewpoint Applied quantum theory to light to explain the photoelectric effect using the concept of light quanta (photons). Albert Einstein en.wikipedia.org/wiki/Photoelectric_effect%23Einstein%27s_explanation (Annalen der Physik)	Mar 1905
The Foundation of General Relativity Published the foundational theory of general relativity, making him internationally famous. Albert Einstein en.wikipedia.org/wiki/General_relativity (Annalen der Physik)	Mar 1916

Skills

Computer Vision: 3D Reconstruction, 3D Gaussian Splatting, Physics-based Rendering, Inverse Rendering

Languages

German

English

Interests

Physics: Quantum Mechanics, Quantum Computing, Quantum Information, Quantum Cryptography, Quantum Communication, Quantum Teleportation

Selected Honors

- Machine Learning Certificate, Stanford University (2018)
- Quantum Computing Certificate, Stanford University (2018)
- Quantum Information Certificate, Stanford University (2018)